

SPHEREx Explorer

The four-minute film, part by part: what is said, what is on screen and what is heard, with a frame from each shot.



3:59.3

Running time (under the 4:00 limit)

**Team
Oblivion**

Bangladesh

v3

Each member named by the narrator

1920 × 1080

30 fps • -14 Lufs

The film at a glance

Four parts, after the NASA Space Apps model for a pitch (“240 seconds of glory”): who, why, what, impact. Times are the finished film's; durations are approximate where shots overlap.

STARTS	PART / SHOT	LENGTH
0:00–0:04	0 • Opening	Who presents
0:00.0	0.1 • The Space Apps title	4.4 s
0:00–1:05	1 • The Dot	Q1 • Who / attention
0:03.0	1.1 • The hook	14.2 s
0:17.2	1.2 • SPHEREx	15.0 s
0:32.2	1.3 • Our work	5.0 s
0:37.2	1.4 • Team Oblivion	15.6 s
0:52.8	1.5 • The name	12.2 s
1:05–2:05	2 • The Blink	Q2 • Why
1:05.0	2.1 • Arizona, 1930	13.0 s
1:18.0	2.2 • Barnard's Star	6.0 s
1:24.0	2.3 • Photograph. Compare. Look for what moved.	12.0 s
1:36.0	2.4 • The scale	14.0 s
1:50.0	2.5 • By hand	15.0 s
2:05–3:03	3 • The demo	Q3 • What
2:05.0	3.1 • A new way to look	3.2 s
2:08.2	3.2 • 01 • Search any place in the sky	10.4 s
2:18.6	3.3 • 02 • Blink the frames	8.2 s
2:26.8	3.4 • 03 • Find what moved • 04 • Check it with NASA JPL	13.8 s
2:40.6	3.5 • 05 • Honest about wavelength	7.2 s
2:47.8	3.6 • 06 • Surveys, a game, Bangla	9.2 s
2:57.0	3.7 • 07 • Ask, and see the sources	6.0 s

3:03–3:59	4 • The Invitation	Q4 • Impact
3:03.0	4.1 • Stay on the discovery	10.9 s
3:13.9	4.2 • Every way of seeing	12.3 s
3:26.2	4.3 • The data, and the people	7.6 s
3:33.8	4.4 • Who it is for	9.6 s
3:43.4	4.5 • Caught	11.0 s
3:54.4	4.6 • End card	4.9 s

Throughout: the long NASA International Space Apps Challenge logo sits in the top-right corner from 0:04 to the final fade. Every sky image is real SPHEREx (or survey) data; every shot of the interface is a recording of the real app on live data. The narrator is a synthetic voice; the music and sound effects are original, generated in code.

WHO PRESENTS

0 • Opening

0:00.0 – 0:04.4

Before the first frame of the story: the challenge, and the team that made the film.



SHOT 0.1

The Space Apps title

0:00.0 – 0:04.4 • 4.4 s

ON SCREEN

Black. The NASA International Space Apps Challenge logo comes into focus out of a soft blur, and a band of light passes across it. Beneath it: **TEAM OBLIVION PRESENTS**. The logo then glides into the top-right corner and becomes the long Space Apps logo, which stays there, small and half-transparent, until the last fade (a broadcaster's corner logo).

NARRATION

No narration.

SOUND

A soft swell into a low impact and a bell chord as the logo resolves; a high bell on “Team Oblivion presents”; a breath as the logo glides away.

1 · The Dot

0:00.0 – 1:05.0

Grab attention with one real moving point, show the scale of SPHEREx, and introduce the people who built the app.



SHOT 1.1

The hook

0:03.0 – 0:17.2 · 14.2 s

ON SCREEN

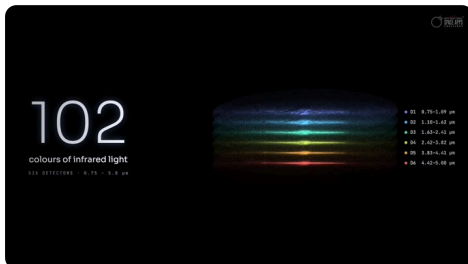
Darkness. Real SPHEREx stars of the field around asteroid (7) Iris appear one by one. One point shifts, back, again: the asteroid, blinking between two real frames. Freeze. On screen: “**Did you see it move?**”

NARRATION

No narration.

SOUND

Silence and air; a small bell as each early star arrives; blips on the blink; a held tone at the freeze.



SHOT 1.2

SPHEREx

0:17.2 – 0:32.2 · 15.0 s

ON SCREEN

The field becomes data: every source ringed, a running count (**1,405 point sources found in this field**), then an estimate for all of the sky in view). The view pulls back. Title: **SPHEREx** · NASA's infrared all-sky survey · launched March 2025. The whole sky in SPHEREx's own colour map, the survey's sweep crossing it; the six detector bands fan out: **102 colours of infrared light** · six detectors · 0.75–5.0 μm . **EVERY SIX MONTHS.**

NARRATION

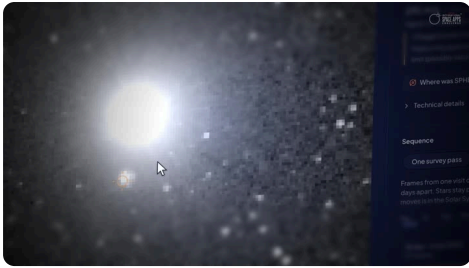
0:19.2 In 2025, NASA launched SPHEREx.

0:22.8 Every six months, it maps the entire sky in 102 colours of infrared light.

0:29.2 Again. And again.

SOUND

The build: pads open, a pulse starts, kick drum; a riser into a hit on the cut.



SHOT 1.3

Our work

0:32.2 – 0:37.2 • 5.0 s

ON SCREEN

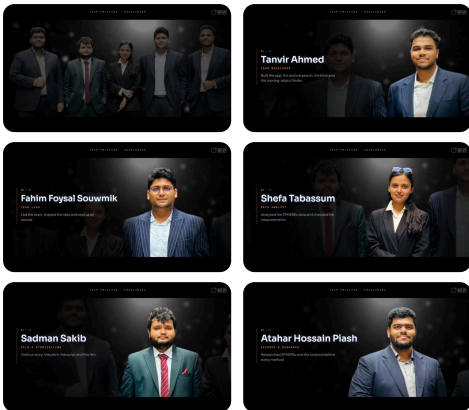
Close and handheld: our own code (the moving-source search), then the app blinking the Iris frames, the pointer on the moving point.

NARRATION

0:32.6 And we couldn't stop asking: what if something important is hiding in there?

SOUND

An impact on the cut; a ticking clock and a quiet piano under the voice.



SHOT 1.4

Team Oblivion

0:37.2 – 0:52.8 • 15.6 s

ON SCREEN

The team rises out of the dark star field (the photo with its background removed). In the top bar: **TEAM OBLIVION · BANGLADESH**. The camera then moves close to each of them in turn, right to left; the others fall into shadow and soft focus. Beside each person, a card: number, name, role and what they did. At the end all five stand in the light together and dissolve into the stars.

THE CARDS, RIGHT TO LEFT

01	Tanvir Ahmed Built the app: the archive search, the blink and the moving-object finder.	LEAD DEVELOPER
02	Fahim Foysal Souwmik Led the team, shaped the idea and kept us on course.	TEAM LEAD
03	Shefa Tabassum Analysed the SPHEREx data and checked the measurements.	DATA ANALYST
04	Sadman Sakib Told our story: the pitch, the script and this film.	FILM & STORYTELLING
05	Atahar Hossain Piash Researched SPHEREx and the science behind every method.	SCIENCE & RESEARCH

NARRATION

0:37.5 We're Team Oblivion, from Bangladesh.
0:40.0 Tanvir Ahmed built the app.
0:42.0 Fahim Foysal Souwmik led the team.
0:44.5 Shefa Tabassum analysed the data.
0:47.0 Sadman Sakib told our story.
0:49.2 And Atahar Hossain Piash researched the science.

SOUND

An impact on the cut; warm strings that change chord as the camera moves; a rising bell as each name is said; a soft pulse; a breath and a chord as all five light up.



SHOT 1.5

The name

0:52.8 - 1:05.0 • 12.2 s

ON SCREEN

The app's front page rises out of the dark. Stars gather into the name: **SPHEREx Explorer**. Beneath it: **“Discovery shouldn't belong only to the people who have the tools to look for it.”** Cut to black.

NARRATION

0:55.4 And we built SPHEREx Explorer because we believe discovery shouldn't belong only to the people who have the tools to look for it.

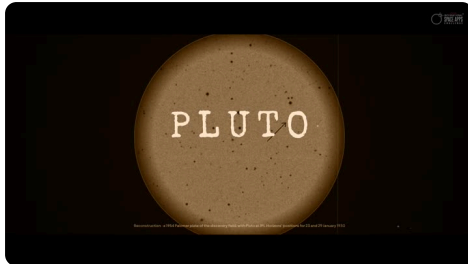
SOUND

A string swell under the name; piano; a reverse swell into a low boom on the cut to black, then a beat of silence.

2 · The Blink

1:05.0 – 2:05.0

Discovery has always meant comparing pictures of the sky by eye. The method is simple; the scale of SPHEREx makes it impossible by hand.



SHOT 2.1

Arizona, 1930

1:05.0 – 1:18.0 · 13.0 s

ON SCREEN

Typed: **LOWELL OBSERVATORY · FLAGSTAFF, ARIZONA / FEBRUARY 1930**. A blink comparator is drawn in; two plates flip back and forth; close on one point; an ink arrow. **PLUTO** · found by Clyde Tombaugh · 18 February 1930. Labelled on screen: a reconstruction (a 1954 Palomar plate of the field, with Pluto at JPL Horizons' positions for 23 and 29 January 1930).

NARRATION

1:06.2 Arizona, 1930. Clyde Tombaugh blinks between two photographs of the same patch of sky...

1:12.8 ...until one tiny point moves.

SOUND

A projector's hum and flutter; the clack of each plate; a typewriter; a low bowed cello; a bell on the point.



SHOT 2.2

Barnard's Star

1:18.0 – 1:24.0 · 6.0 s

ON SCREEN

Barnard's Star · the same 24' of sky in five surveys: Palomar plates from the 1950s and 1991, 2MASS, AllWISE and SPHEREx in 2026. The star visibly moves across 75 years.

NARRATION

1:18.4 Almost a century later, the method hasn't changed.

SOUND

A tick on each survey.



SHOT 2.3

Photograph. Compare. Look for what moved.

1:24.0 - 1:36.0 · 12.0 s

ON SCREEN

Photograph · Compare · Look for what moved: three real SPHEREx frames of Iris, side by side; they blink. Then one frame among a wall of SPHEREx images, the view pulling back.

NARRATION

1:24.0 Photograph the sky. Compare. Look for what moved.

SOUND

A shutter on each frame; blips on the blink; the music builds.



SHOT 2.4

The scale

1:36.0 - 1:50.0 · 14.0 s

ON SCREEN

Hush on the wall of images. Then it falls away into points, and the numbers stack up: **102 colours x the whole sky x every six months · 450 million galaxies · 100 million stars in the Milky Way** (NASA's expectations for the SPHEREx survey).

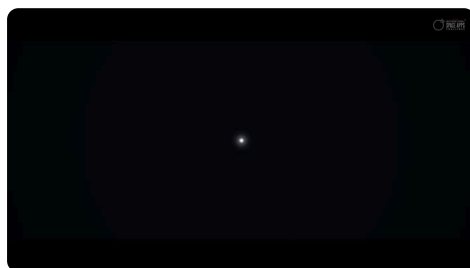
NARRATION

1:36.0 The method is simple.

1:39.4 The scale isn't.

SOUND

Silence on “The method is simple”; then a driving minor progression, louder and louder, and a riser.



SHOT 2.5

By hand

1:50.0 - 2:05.0 · 15.0 s

ON SCREEN

Someone scrolling an endless archive of SPHEREx images, stopping to compare by eye, scrolling on. Too many points to count. One remains: a single dot. Small title: **SPHEREx EXPLORER**.

NARRATION

1:50.6 Somewhere inside all of this, something could be moving.

1:56.6 The question is: will anyone notice?

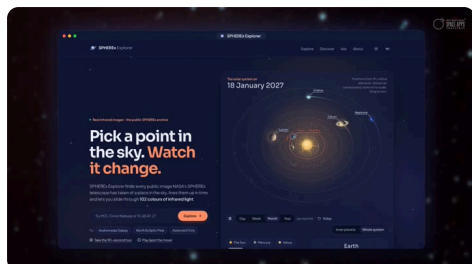
SOUND

Sparse piano and a tension that never resolves; everything stops on the dot.

3 • The demo

2:05.0 – 3:03.0

Real screen recordings of SPHEREx Explorer on live data. A numbered step label sits top left.



SHOT 3.1

A new way to look

2:05.0 – 2:08.2 • 3.2 s

ON SCREEN

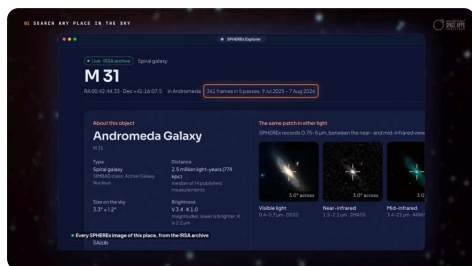
The dot opens into the app.

NARRATION

2:05.8 So we built a new way to look.

SOUND

A whoosh up and a bell.



SHOT 3.2

01 • Search any place in the sky

2:08.2 – 2:18.6 • 10.4 s

ON SCREEN

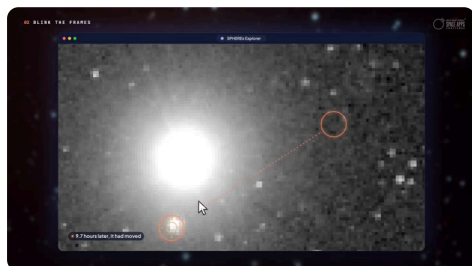
“Andromeda Galaxy” typed into the search; the app finds every SPHEREx frame of it (341 frames in 5 passes) and the timeline plays. Chip: **Every SPHEREx image of this place, from the IRSA archive.**

NARRATION

2:09.4 Name any place in the sky. SPHEREx Explorer finds every image SPHEREx has taken of it, live from the public archive.

SOUND

Clicks and keys from the recording; a steady pulse.



SHOT 3.3

02 • Blink the frames

2:18.6 – 2:26.8 • 8.2 s

ON SCREEN

The Iris case blinking in the app; where the point was, and where it went. Chip: **9.7 hours later, it had moved.**

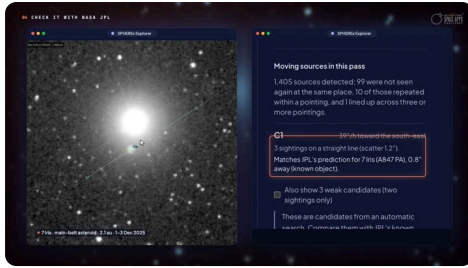
NARRATION

2:18.6 Then it does what Tombaugh did. It blinks the frames.

2:23.4 And something moves.

SOUND

UI sounds from the recording.



SHOT 3.4

03 · Find what moved · 04 · Check it with NASA JPL

2:26.8 – 2:40.6 · 13.8 s

ON SCREEN

The app's own search finds candidate C1 (**Our own search: knows nothing about asteroids**). Check JPL draws the predicted track (**JPL's predicted track: the same object**): **Matches JPL's prediction for 7 Iris, 0.8" away**. Chip: 7 Iris · main-belt asteroid · 2.1 au · 1–3 Dec 2025.

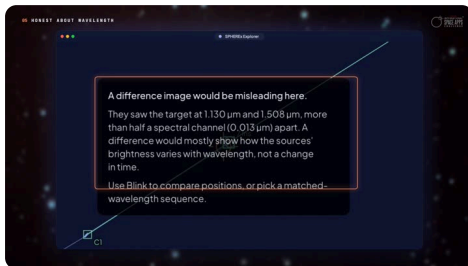
NARRATION

2:27.0 Its own search finds the moving point, knowing nothing about asteroids. Then it checks with NASA's JPL.

2:34.6 It's the asteroid Iris, caught by SPHEREx in December 2025.

SOUND

UI sounds; a soft lock-on.



SHOT 3.5

05 · Honest about wavelength

2:40.6 – 2:47.8 · 7.2 s

ON SCREEN

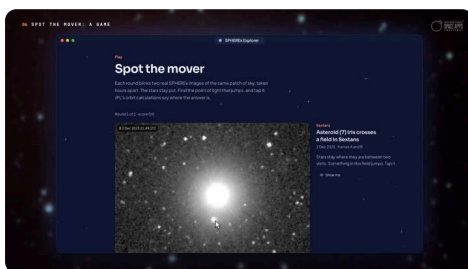
Each frame saw the target at a different wavelength, so the app refuses a misleading comparison: **“A difference image would be misleading here.”**

NARRATION

2:40.6 And it's honest. Each frame sees a different colour, so it never mistakes colour for change.

SOUND

The pulse continues under the voice.



SHOT 3.6

06 · Surveys, a game, Bangla

2:47.8 – 2:57.0 · 9.2 s

ON SCREEN

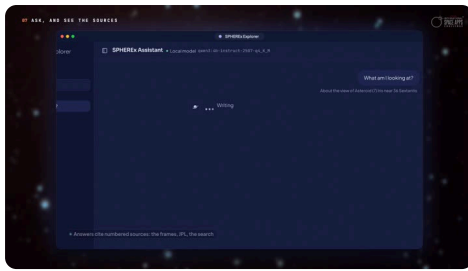
Three quick panels: **75 YEARS OF SURVEYS, SIDE BY SIDE** (the decades blink); **SPOT THE MOVER: A GAME; ENGLISH · বাংলা** (the Bangla switch).

NARRATION

2:48.0 Seventy-five years of sky surveys, side by side. A game that trains your eye. In English, and in Bangla.

SOUND

A soft swish on each panel.



SHOT 3.7

07 • Ask, and see the sources

2:57.0 - 3:03.0 • 6.0 s

ON SCREEN

The assistant answers a question about the view, with numbered sources. Chip: **Answers cite numbered sources: the frames, JPL, the search.** Hard cut.

NARRATION

2:57.4 Ask it anything. Every answer shows its sources.

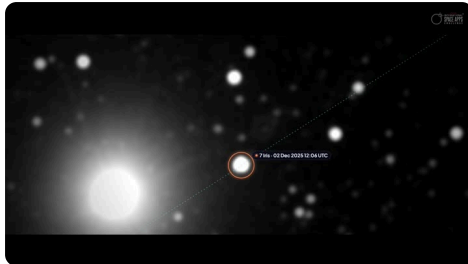
SOUND

Keys as the question is typed; the music cuts with the picture.

4 • The Invitation

3:03.0 – 3:59.3

From one asteroid to the whole of human seeing: more data than ever, and the people it needs. The first shot's dot, caught at last.



SHOT 4.1

Stay on the discovery

3:03.0 – 3:13.9 • 10.9 s

ON SCREEN

Iris stepping along its track, ringed: **7 Iris • 02 Dec 2025 • UTC**. The view pulls back to the Solar System on 2 December 2025 (orbits from JPL), then to Earth, Bangladesh toward us.

NARRATION

3:10.4 For most of human history, the universe was something we could only look at.

SOUND

Blips on each step; a whoosh out; a quiet pad and piano.



SHOT 4.2

Every way of seeing

3:13.9 – 3:26.2 • 12.3 s

ON SCREEN

Drawn one by one: **THE NAKED EYE** (thousands of years ago), **THE TELESCOPE** (1609), **PHOTOGRAPHIC PLATES** (1930), **THE GREAT OBSERVATORIES** (1949), **SPHEREx** (2025); in a row; then SPHEREx alone.

NARRATION

3:20.8 Every generation built a better way to see.

3:24.4 Our generation has something different.

SOUND

A pencil sound on each drawing; a building chord progression.



SHOT 4.3

The data, and the people

3:26.2 – 3:33.8 • 7.6 s

ON SCREEN

The whole sky unfolds from SPHEREx: **More data than ever before.** Hundreds of small rings light up across it: **Now we need more people looking at it.**

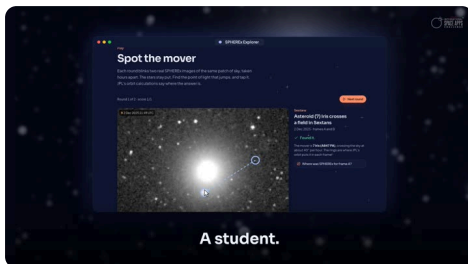
NARRATION

3:27.8 We have more data than ever before.

3:31.2 Now we need more people looking at it.

SOUND

An unfolding whoosh; many small bells as the rings appear.



SHOT 4.4

Who it is for

3:33.8 – 3:43.4 • 9.6 s

ON SCREEN

A student. (the game: “Found it.”) · **A citizen scientist.** (Discover) · **An astronomer.** (JPL's match) · **Anyone curious enough to look.** (fast cuts through the app, including **In Bangla, on a phone**). Cut to black.

NARRATION

3:35.4 A student.

3:37.2 A citizen scientist.

3:39.2 An astronomer.

3:41.0 Anyone curious enough to look.

SOUND

Driving music with a hit on each cut; a boom on the cut to black.



SHOT 4.5

Caught

3:43.4 – 3:54.4 • 11.0 s

ON SCREEN

The first shot's field again: “**Did you see it move?**” This time it is caught: **Moving · 53" an hour, Matches JPL: 7 Iris.** The view pulls back to the whole sky.

NARRATION

3:46.4 Because the next discovery may not be hiding from humanity.

3:50.8 It may simply be waiting for someone to notice.

SOUND

Blips and a lock-on; a whoosh out; a riser into the final chord.



SHOT 4.6

End card

3:54.4 – 3:59.3 • 4.9 s

ON SCREEN

SPHEREx Explorer · Look closer. Discover what changed. · Team Oblivion · Bangladesh · NASA Space Apps Challenge 2026. Credits in small type: SPHEREx data NASA/JPL–Caltech/IPAC via IRSA · SPHEREx QR2 maps via CDS hips2fits · DSS (STScI/AURA, Palomar) · 2MASS · AllWISE · JPL Horizons & SBIdent · Narration: synthetic voice · Music: original · not affiliated with or endorsed by NASA, JPL, Caltech or IPAC. Fade to black.

NARRATION

No narration.

SOUND

The final chord, strings, pad and piano, ringing out; fade.

Versions: **SPHEREx–Explorer–film.mp4** (narrated) · **–captioned.mp4** (English captions burned in) · **–no-voice.mp4** (music and effects only, for narrating live) · **v2/** (the same film with the team cards unspoken). Narration timings come from the film's own files (script.json, edit.json), so they match the video to the tenth of a second.